

2024, Mar. 2024 Release LOC:Acute Adult
Hematology/Oncology: Hemolytic Uremic Syndrome
Utilization Benchmarks: *Multiple options available, see table below for options*

Overview
Select Day
 Episode Day 1
 Episode Day 2
 Episode Day 3
 Episode Day 4
 Episode Day 5-8
 Episode Day 9
Discharge Screens ⁽³⁵⁾
Utilization Benchmarks
Notes

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Overview

Introduction

Hemolytic uremic syndrome (HUS) is a disease that occurs as a result of food born illnesses or contaminated water supply. The most common contaminants include shiga-toxin-producing *E. coli* and *shigella* and is commonly found in pediatric patients but can also affect adults.

Evaluation and treatment

Hemolytic uremic syndrome is usually preceded by infectious and/or bloody diarrhea and develops 5 to 10 days after the onset of diarrhea. Symptoms include decreased urine output, hematuria, abdominal pain, nausea or vomiting, diarrhea, hypertension, edema, fatigue, and confusion. Laboratory abnormalities include thrombocytopenia, low platelet count, increased lactate dehydrogenase, elevated creatinine, decreased haptoglobin, and microangiopathic hemolytic anemia. Complications of hemolytic uremic syndrome include acute kidney injury, hypertension, pancreatitis, stroke, myocardial infarction, encephalopathy, seizure, coma, or death. Treatment for HUS is predominantly supportive care. There is insufficient evidence to support the use of anti-infectives and these medications may potentially exacerbate symptoms due to the increased release of toxins. (Karpman et al., J Intern Med 2017, 281: 123-48)

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, Agency for Healthcare Research and Quality (AHRQ) Comparative Effectiveness Reviews, the National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the *Cochrane Handbook*. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The content is based on a variety of references which are cited at specific criteria points throughout the subset.

(Excludes PO medications unless noted)

Select Day, One:**● Episode Day 1, One:**

(Symptom or finding within 24h)

● ACUTE, All: ⁽¹⁾**● Finding, ≥ Two:****● Elevated liver enzyme, ≥ One:**

– ALT > 2x ULN

– AST > 2x ULN

– Creatinine > 3.0 mg/dL(265.2 µmol/L)

– Hct < 18%(0.18) or Hb < 6.0 g/dL(60 g/L)

● Proteinuria, ≥ One:

– ≥ 300 mg in a 24h urine sample

● > 2 random urine samples last 24h, ≥ One:

– ≥ 30 mg/dL

– ≥ 2+

– Hct or Hb monitoring at least daily

● Treatment, ≥ One:– Apheresis ⁽²⁾**● Dialysis (initial) and ≤ 7d since initiation, One:** ⁽³⁾

– Hemodialysis

– Peritoneal

– Fresh frozen plasma (FFP)

– Immunosuppressant ⁽⁴⁾**● INTERMEDIATE, ≥ One:** ⁽⁵⁾**● Hyperkalemia, All:**

– Potassium ≥ 5.6 mEq/L(5.6 mmol/L)

● Electrocardiogram (ECG) changes, ≥ One: ⁽⁶⁾

– Peaked T waves

– PVCs > 6/min

● Intervention, ≥ One:

– Calcium chloride or gluconate

– Glucose 50%(0.50) with insulin ⁽⁷⁾**● Seizure and, Both:**

– Activity resolved

– Anticonvulsant, continuous

● Oxygen, ≥ One: ⁽⁸⁾– HFNC and respiratory monitoring every 3-4h ^(9, 10)

– Oxygen ≥ 40%(0.40) ≤ 2d

● CRITICAL, ≥ One: ⁽¹¹⁾**● Hemolytic uremic syndrome and, Both:****● Finding, ≥ One:****● Potassium > 6.0 mEq/L(6.0 mmol/L) and, ≥ One:**

– Loss of P wave

– Multifocal PVCs

– Neuromuscular deficit ⁽¹²⁾

– Widening QRS

● Sodium, One:

– < 120 mEq/L(120 mmol/L) and volume overload

– > 160 mEq/L(160 mmol/L)

● Intervention, ≥ One:

– Calcium chloride or gluconate

– CRRT ⁽¹³⁾

- IV fluid resuscitation $\leq 2d$ ⁽¹⁴⁾
- Diuretic, continuous
- **Dialysis, One:**
 - Hemodialysis $\leq 7d$ since initiation
 - Peritoneal dialysis $\leq 7d$ since initiation
 - Glucose 50%(0.50) with insulin ⁽⁷⁾
- **Anticonvulsant and, $\geq$ One:**
 - Recurrent seizure ⁽¹⁵⁾
 - Refractory seizure ⁽¹⁵⁾
- Coma, stupor, obtundation or GCS ≤ 8 and neurological assessment every 1-2h, $\leq 2d$ ^(16, 17)
- Mechanical ventilation
- **NIPPV and, $\geq$ One:** ⁽¹⁸⁾
 - Arterial $PO_2 \leq 39$ mmHg(5.2 kPa)
- **Arterial or venous $P_{CO_2} \geq 60$ mmHg(8.0 kPa) and, One:**
 - COPD and arterial or venous pH ≤ 7.24
 - No hx of COPD
- **Oxygen, Both:** ⁽⁸⁾
 - HFNC ⁽⁹⁾
 - Respiratory monitoring every 1-2h ⁽¹⁰⁾
- **Episode Day 2, One:**
 - **ACUTE, Both:**
 - Hct or Hb monitoring at least daily
 - **Treatment, $\geq$ One:**
 - Apheresis ⁽²⁾
 - **Dialysis (initial) and $\leq 7d$ since initiation, One:** ⁽³⁾
 - Hemodialysis
 - Peritoneal
 - Fresh frozen plasma (FFP)
 - Immunosuppressant ⁽⁴⁾
- **INTERMEDIATE, $\geq$ One:**
 - **Hyperkalemia, All:**
 - Potassium ≥ 5.6 mEq/L(5.6 mmol/L)
 - **Electrocardiogram (ECG) changes, $\geq$ One:** ⁽⁶⁾
 - Peaked T waves
 - PVCs $> 6/min$
 - **Intervention, $\geq$ One:**
 - Calcium chloride or gluconate
 - Glucose 50%(0.50) with insulin ⁽⁷⁾
 - NIPPV ⁽¹⁸⁾
 - **Oxygen, $\geq$ One:** ⁽⁸⁾
 - HFNC and respiratory monitoring every 3-4h ^(9, 10)
 - Oxygen $\geq 40\%(0.40) \leq 2d$
 - **Seizure and, Both:**
 - Activity resolved
 - Anticonvulsant continuous
- **CRITICAL, $\geq$ One:**
 - **Hemolytic uremic syndrome and, Both:**
 - **Finding, $\geq$ One:**
 - **Potassium > 6.0 mEq/L(6.0 mmol/L) and, $\geq$ One:**
 - Loss of P wave
 - Multifocal PVCs
 - Neuromuscular deficit ⁽¹²⁾
 - Widening QRS

- **Sodium, One:**
 - < 120 mEq/L(120 mmol/L) and volume overload
 - > 160 mEq/L(160 mmol/L)
- **Intervention, ≥ One:**
 - Calcium chloride or gluconate
 - CRRT ⁽¹³⁾
 - IV fluid resuscitation ≤ 2d ⁽¹⁴⁾
 - Diuretic, continuous
- **Dialysis, One:**
 - Hemodialysis ≤ 7d since initiation
 - Peritoneal dialysis ≤ 7d since initiation
 - Glucose 50%(0.50) with insulin ⁽⁷⁾
- **Anticonvulsant and, ≥ One:**
 - Recurrent seizure ⁽¹⁵⁾
 - Refractory seizure ⁽¹⁵⁾
- Coma, stupor, obtundation or GCS ≤ 8 and neurological assessment every 1-2h, ≤ 2d ^(16, 17)
- Mechanical ventilation
- **NIPPV and, ≥ One:** ⁽¹⁸⁾
 - FiO₂ > 50%(0.50)
 - Respiratory intervention every 1-2h ⁽¹⁹⁾
- HFNC and respiratory monitoring every 1-2h ^(8, 9, 10)
- **Episode Day 3, One:**
 - **ACUTE, ≥ One:**
 - Apheresis ⁽²⁾
 - **Dialysis (initial) and ≤ 7d since initiation, One:** ⁽³⁾
 - Hemodialysis
 - Peritoneal
 - Dialysis discontinued and renal function monitoring ≤ 2d since discontinuation
 - Fresh frozen plasma (FFP)
 - Hct or Hb monitoring at least daily, ≤ 2d
 - Immunosuppressant ⁽⁴⁾
- **INTERMEDIATE, ≥ One:**
 - **Hyperkalemia, All:**
 - Potassium ≥ 5.6 mEq/L(5.6 mmol/L)
 - **Electrocardiogram (ECG) changes, ≥ One:** ⁽⁶⁾
 - Peaked T waves
 - PVCs > 6/min
 - **Intervention, ≥ One:**
 - Calcium chloride or gluconate
 - Glucose 50%(0.50) with insulin ⁽⁷⁾
- NIPPV ⁽¹⁸⁾
- **Oxygen, ≥ One:** ⁽⁸⁾
 - HFNC and respiratory monitoring every 3-4h ^(9, 10)
 - Oxygen ≥ 40%(0.40) ≤ 2d
- **Seizure and, Both:**
 - Activity resolved
 - Anticonvulsant continuous
- **CRITICAL, ≥ One:** ⁽¹¹⁾
 - **Hemolytic uremic syndrome and, Both:**
 - **Finding, ≥ One:**
 - **Potassium > 6.0 mEq/L(6.0 mmol/L) and, ≥ One:**
 - Loss of P wave
 - Multifocal PVCs

- Neuromuscular deficit ⁽¹²⁾
- Widening QRS
- Sodium, **One**:
 - < 120 mEq/L(120 mmol/L) and volume overload
 - > 160 mEq/L(160 mmol/L)
- Intervention, ≥ **One**:
 - Calcium chloride or gluconate
 - CRRT ⁽¹³⁾
 - IV fluid resuscitation ≤ 2d ⁽¹⁴⁾
 - Diuretic, continuous
- Dialysis, **One**:
 - Hemodialysis ≤ 7d since initiation
 - Peritoneal dialysis ≤ 7d since initiation
 - Glucose 50%(0.50) with insulin ⁽⁷⁾
- Anticonvulsant and, ≥ **One**:
 - Recurrent seizure ⁽¹⁵⁾
 - Refractory seizure ⁽¹⁵⁾
- Coma, stupor, obtundation or GCS ≤ 8 and neurological assessment every 1-2h, ≤ 2d ^(16, 17)
- Mechanical ventilation
- NIPPV and, ≥ **One**: ⁽¹⁸⁾
 - FiO₂ > 50%(0.50)
 - Respiratory intervention every 1-2h ⁽¹⁹⁾
- HFNC and respiratory monitoring every 1-2h ^(8, 9, 10)
- Episode Day 4, **One**:
 - ACUTE, **One**:
 - Early Responder, discharge expected today if clinically stable last 12h, **All**: ⁽²⁰⁾
 - Afebrile
 - Kidney function improving or within acceptable limits ⁽²¹⁾
 - Lab values within acceptable limits ⁽²¹⁾
 - Outpatient dialysis regimen, **One**:
 - Established
 - Not indicated
 - Tolerating PO ⁽²²⁾
 - Complication or comorbidity, **One**:
 - No complication or active comorbidity relevant to this episode of care ⁽²³⁾
 - Complication or comorbidity and clinically stable for discharge, ≥ **One**:
 - Blood pressure, **All**:
 - Antihypertensive regimen established
 - Systolic 90-160 mmHg or within acceptable limits ^(21, 24)
 - Diastolic ≤ 100 mmHg or within acceptable limits ^(21, 24)
 - Blood glucose 70-250 mg/dL(3.9-13.9 mmol/L) or within acceptable limits ⁽²¹⁾
 - O₂ sat ≥ 92%(0.92) or within acceptable limits ⁽²¹⁾
 - Electrolytes within acceptable limits ⁽²¹⁾
 - Partial responder, not clinically stable for discharge and requires continued stay, ≥ **One**:
 - Hemolytic uremic syndrome and, ≥ **One**:
 - Apheresis ⁽²⁾
 - Dialysis (initial) and ≤ 7d since initiation, **One**: ⁽³⁾
 - Hemodialysis
 - Peritoneal
 - Dialysis discontinued and renal function monitoring ≤ 2d since discontinuation
 - Fresh frozen plasma (FFP)
 - Hct or Hb monitoring at least daily, ≤ 2d
 - Immunosuppressant ⁽⁴⁾

- Blood glucose $\leq$ 24h, **All:** ⁽²⁵⁾
 - Finding, $\geq$ **One:**
 - > 250 mg/dL(13.9 mmol/L)
 - < 70 mg/dL(3.9 mmol/L)
 - Blood glucose monitoring at least 4x/24h
- Insulin adjustment and, **One:** ^(25, 26)
 - IV ≥ 2 x/24h
 - SC ≥ 4 x/24h
- Fever, new onset and, **Both:** ⁽²⁷⁾
 - Temperature $\geq 100.4^{\circ}\text{F}(38.0^{\circ}\text{C})$ ⁽²⁸⁾
- Intervention, $\geq$ **One:**
 - Culture pending ≤ 2 d
 - Drug-induced fever suspected and precipitating drug discontinued ≤ 3 d ⁽²⁹⁾
 - Imaging study ≤ 24 h
- Hyperkalemia, **All:**
 - Potassium > 5.5 mEq/L(5.5 mmol/L)
 - No electrocardiogram (ECG) changes ^(30, 31)
- Intervention, $\geq$ **One:**
 - Calcium chloride or gluconate
 - Diuretic (includes PO)
 - Glucose 50%(0.50) with insulin ⁽⁷⁾
 - IV fluid ⁽³²⁾
 - Potassium binding agent (includes PO or PR) ⁽³³⁾
- Hypertension, **Both:**
 - Blood pressure, $\geq$ **One:**
 - Systolic > 160 mmHg, and $>$ baseline ⁽³⁴⁾
 - Diastolic > 100 mmHg, and $>$ baseline ⁽³⁴⁾
 - Hypertensive medication adjustment (includes PO) requiring change in medication, dose, or frequency ≤ 2 d
- Oxygenation, **One:**
 - O_2 sat $\geq 92\%(0.92)$ or baseline ⁽³⁴⁾
 - O_2 sat $\leq 91\%(0.91)$ and $<$ baseline requiring supplemental oxygen ^(8, 34)
 - Post Critical care ≤ 24 h
- INTERMEDIATE, $\geq$ **One:**
 - Hyperkalemia, **All:** ⁽⁶⁾
 - Potassium ≥ 5.6 mEq/L(5.6 mmol/L)
 - Electrocardiogram (ECG) changes, $\geq$ **One:** ⁽⁶⁾
 - Peaked T waves
 - PVCs $> 6/\text{min}$
 - Intervention, $\geq$ **One:**
 - Calcium chloride or gluconate
 - Glucose 50%(0.50) with insulin ⁽⁷⁾
 - NIPPV ⁽¹⁸⁾
 - Oxygen, $\geq$ **One:** ⁽⁸⁾
 - HFNC and respiratory monitoring every 3-4h ^(9, 10)
 - Oxygen $\geq 40\%(0.40) \leq 2$ d
 - Seizure and, **Both:**
 - Activity resolved
 - Anticonvulsant continuous
- CRITICAL, $\geq$ **One:**
 - Hemolytic uremic syndrome and, **Both:**
 - Finding, $\geq$ **One:**
 - Potassium > 6.0 mEq/L(6.0 mmol/L) and, $\geq$ **One:**

- Loss of P wave
- Multifocal PVCs
- Neuromuscular deficit ⁽¹²⁾
- Widening QRS
- **Sodium, One:**
 - < 120 mEq/L(120 mmol/L) and volume overload
 - > 160 mEq/L(160 mmol/L)
- **Intervention, ≥ One:**
 - Calcium chloride or gluconate
 - CRRT ⁽¹³⁾
 - IV fluid resuscitation ≤ 2d ⁽¹⁴⁾
 - Diuretic, continuous
- **Dialysis, One:**
 - Hemodialysis ≤ 7d since initiation
 - Peritoneal dialysis ≤ 7d since initiation
 - Glucose 50%(0.50) with insulin ⁽⁷⁾
- **Anticonvulsant and, ≥ One:**
 - Recurrent seizure ⁽¹⁵⁾
 - Refractory seizure ⁽¹⁵⁾
- Coma, stupor, obtundation or GCS ≤ 8 and neurological assessment every 1-2h, ≤ 2d ^(16, 17)
- Mechanical ventilation
- **NIPPV and, ≥ One:** ⁽¹⁸⁾
 - FiO₂ > 50%(0.50)
 - Respiratory intervention every 1-2h ⁽¹⁹⁾
- HFNC and respiratory monitoring every 1-2h ^(8, 9, 10)
- **Episode Day 5-8, One:**
 - **ACUTE, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽²⁰⁾
 - Afebrile
 - Kidney function improving or within acceptable limits ⁽²¹⁾
 - Lab values within acceptable limits ⁽²¹⁾
 - **Outpatient dialysis regimen, One:**
 - Established
 - Not indicated
 - Tolerating PO ⁽²²⁾
 - **Complication or comorbidity, One:**
 - No complication or active comorbidity relevant to this episode of care ⁽²³⁾
 - **Complication or comorbidity and clinically stable for discharge, ≥ One:**
 - **Blood pressure, All:**
 - Antihypertensive regimen established
 - Systolic 90-160 mmHg or within acceptable limits ^(21, 24)
 - Diastolic ≤ 100 mmHg or within acceptable limits ^(21, 24)
 - Blood glucose 70-250 mg/dL(3.9-13.9 mmol/L) or within acceptable limits ⁽²¹⁾
 - O₂ sat ≥ 92%(0.92) or within acceptable limits ⁽²¹⁾
 - Electrolytes within acceptable limits ⁽²¹⁾
- **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
 - **Hemolytic uremic syndrome and, ≥ One:**
 - Apheresis ⁽²⁾
 - **Dialysis (initial) and ≤ 7d since initiation, One:** ⁽³⁾
 - Hemodialysis
 - Peritoneal
 - Dialysis discontinued and renal function monitoring ≤ 2d since discontinuation
 - Fresh frozen plasma (FFP)

- Hct or Hb monitoring at least daily, $\leq 2d$
- Immunosuppressant ⁽⁴⁾
- Blood glucose $\leq 24h$, **All:** ⁽²⁵⁾
 - Finding, $\geq$ **One:**
 - > 250 mg/dL(13.9 mmol/L)
 - < 70 mg/dL(3.9 mmol/L)
 - Blood glucose monitoring at least 4x/24h
 - Insulin adjustment and, **One:** ^(25, 26)
 - IV $\geq 2x/24h$
 - SC $\geq 4x/24h$
- Fever, new onset and, **Both:** ⁽²⁷⁾
 - Temperature $\geq 100.4^{\circ}F(38.0^{\circ}C)$ ⁽²⁸⁾
 - Intervention, $\geq$ **One:**
 - Culture pending $\leq 2d$
 - Drug-induced fever suspected and precipitating drug discontinued $\leq 3d$ ⁽²⁹⁾
 - Imaging study $\leq 24h$
- Hyperkalemia, **All:**
 - Potassium > 5.5 mEq/L(5.5 mmol/L)
 - No electrocardiogram (ECG) changes ^(30, 31)
 - Intervention, $\geq$ **One:**
 - Calcium chloride or gluconate
 - Diuretic (includes PO)
 - Glucose 50%(0.50) with insulin ⁽⁷⁾
 - IV fluid ⁽³²⁾
 - Potassium binding agent (includes PO or PR) ⁽³³⁾
- Hypertension, **Both:**
 - Blood pressure, $\geq$ **One:**
 - Systolic > 160 mmHg, and $>$ baseline ⁽³⁴⁾
 - Diastolic > 100 mmHg, and $>$ baseline ⁽³⁴⁾
 - Hypertensive medication adjustment (includes PO) requiring change in medication, dose, or frequency $\leq 2d$
- Oxygenation, **One:**
 - O_2 sat $\geq 92\%(0.92)$ or baseline ⁽³⁴⁾
 - O_2 sat $\leq 91\%(0.91)$ and $<$ baseline requiring supplemental oxygen ^(8, 34)
 - Post Critical care $\leq 24h$
- INTERMEDIATE, $\geq$ **One:**
 - Hyperkalemia, **All:**
 - Potassium ≥ 5.6 mEq/L(5.6 mmol/L)
 - Electrocardiogram (ECG) changes, $\geq$ **One:** ⁽⁶⁾
 - Peaked T waves
 - PVCs $> 6/min$
 - Intervention, $\geq$ **One:**
 - Calcium chloride or gluconate
 - Glucose 50%(0.50) with insulin ⁽⁷⁾
- NIPPV ⁽¹⁸⁾
- Oxygen, $\geq$ **One:** ⁽⁸⁾
 - HFNC and respiratory monitoring every 3-4h ^(9, 10)
 - Oxygen $\geq 40\%(0.40)$ $\leq 2d$
- Seizure and, **Both:**
 - Activity resolved
 - Anticonvulsant continuous
- CRITICAL, $\geq$ **One:**
 - Hemolytic uremic syndrome and, **Both:**

- Finding, ≥ **One**:
 - Potassium > 6.0 mEq/L(6.0 mmol/L) and, ≥ **One**:
 - Loss of P wave
 - Multifocal PVCs
 - Neuromuscular deficit ⁽¹²⁾
 - Widening QRS
 - Sodium, **One**:
 - < 120 mEq/L(120 mmol/L) and volume overload
 - > 160 mEq/L(160 mmol/L)
- Intervention, ≥ **One**:
 - Calcium chloride or gluconate
 - CRRT ⁽¹³⁾
 - IV fluid resuscitation ≤ 2d ⁽¹⁴⁾
 - Diuretic, continuous
 - Dialysis, **One**:
 - Hemodialysis ≤ 7d since initiation
 - Peritoneal dialysis ≤ 7d since initiation
 - Glucose 50%(0.50) with insulin ⁽⁷⁾
- Anticonvulsant and, ≥ **One**:
 - Recurrent seizure ⁽¹⁵⁾
 - Refractory seizure ⁽¹⁵⁾
- Coma, stupor, obtundation or GCS ≤ 8 and neurological assessment every 1-2h, ≤ 2d ^(16, 17)
- Mechanical ventilation
- NIPPV and, ≥ **One**: ⁽¹⁸⁾
 - FiO₂ > 50%(0.50)
 - Respiratory intervention every 1-2h ⁽¹⁹⁾
- HFNC and respiratory monitoring every 1-2h ^(8, 9, 10)
- Episode Day 9, **One**:
 - ACUTE, **One**:
 - Responder, discharge expected today if clinically stable last 12h, **All**: ⁽²⁰⁾
 - Afebrile
 - Kidney function improving or within acceptable limits ⁽²¹⁾
 - Lab values within acceptable limits ⁽²¹⁾
 - Outpatient dialysis regimen, **One**:
 - Established
 - Not indicated
 - Tolerating PO ⁽²²⁾
 - Complication or comorbidity, **One**:
 - No complication or active comorbidity relevant to this episode of care ⁽²³⁾
 - Complication or comorbidity and clinically stable for discharge, ≥ **One**:
 - Blood pressure, **All**:
 - Antihypertensive regimen established
 - Systolic 90-160 mmHg or within acceptable limits ^(21, 24)
 - Diastolic ≤ 100 mmHg or within acceptable limits ^(21, 24)
 - Blood glucose 70-250 mg/dL(3.9-13.9 mmol/L) or within acceptable limits ⁽²¹⁾
 - O₂ sat ≥ 92%(0.92) or within acceptable limits ⁽²¹⁾
 - Electrolytes within acceptable limits ⁽²¹⁾
 - Non-responder, not clinically stable for discharge and exceeds episode days at the Acute level of care for this condition, **One**:
 - Use Extended Stay subset
 - Refer for secondary review
- INTERMEDIATE, **One**:
 - Non-responder, not clinically stable for discharge and exceeds episode days at the Intermediate level of

care for this condition, **One:**

- Use **Extended Stay subset**
- Refer for secondary review

● **CRITICAL, One:**

● **Non-responder**, not clinically stable for discharge and exceeds episode days at the Critical level of care for this condition, **One:**

- Use **Extended Stay subset**
- Refer for secondary review

Discharge, Level of Care, One: ⁽³⁵⁾

● **HOME, Both:**

● Level of care appropriateness, **Both:**

- Home environment safe and accessible ⁽³⁶⁾
- Patient or caregiver demonstrates ability to manage care

● Discharge planning, **All:**

- Provide comprehensive written discharge and teaching instructions ⁽³⁷⁾
- Medication reconciliation ⁽³⁸⁾
- Follow-up care planned ⁽³⁹⁾
- Identify and address transportation needs

● **HOME CARE, Both:**

● Level of care appropriateness, **All:**

- Home environment safe and accessible ⁽³⁶⁾
- Patient or caregiver willing and able to learn care
- Outpatient management contraindicated or unavailable ^(40, 41)

● Skilled services, ≥ **One:** ⁽⁴²⁾

- Skilled nursing ⁽⁴³⁾
- Skilled therapy, PT, OT, or ST
- Behavioral health ⁽⁴⁴⁾

● Discharge planning, **All:**

- Provide comprehensive written discharge and teaching instructions ⁽³⁷⁾
- Medication reconciliation ⁽³⁸⁾
- Follow-up care planned ⁽³⁹⁾
- Identify and address transportation needs

● **BEHAVIORAL HEALTH, Both:**

● Level of care appropriateness, **One:**

- Support systems available ⁽⁴⁵⁾

● Skilled treatment, ≥ **One:**

- Clinical assessment ⁽⁴⁶⁾
- Individual, group, or family counseling
- Psychiatric, substance, or medication evaluation

● **SKILLED MEDICAL OR THERAPY, Both:**

● Level of care appropriateness, **All:**

- Medical practitioner, NP, PA assessment or oversight ≥ 1x/wk
- Treatment precluded in a lower level

● Services, ≥ **One:**

- Medical and skilled nursing interventions or assessment 1-2x/24h ⁽⁴³⁾

● Therapy, **All:**

- Goal directed therapy with rehabilitation potential ^(47, 48)
- Functional impairments requiring at least supervision ⁽⁴⁹⁾
- Able to tolerate 1h/d of skilled therapy ≥ 5d/wk

● Complete prior to facility transfer, **All:**

- Medication reconciliation ⁽³⁸⁾
- Obtain and complete forms for facility

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- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● SUBACUTE MEDICAL OR THERAPY, Both:

- Level of care appropriateness, **All:**
 - Medical practitioner, NP, PA assessment or oversight $\geq 2x/wk$
 - Treatment precluded in a lower level
- Services, $\geq$ **One:**
 - Medical and skilled nursing interventions or assessment 4h/24h ⁽⁴³⁾
- Therapy, **All:**
 - Goal directed therapy with rehabilitation potential ^(47, 48)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽⁴⁹⁾
 - Able to tolerate 2-3h/d of skilled therapy $\geq 5d/wk$ and ≥ 1 therapy discipline
- Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽³⁸⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation

● SUBACUTE REHAB, Both:

- Level of care appropriateness, **All:**
 - Medical practitioner, NP, PA assessment or oversight $\geq 3x/wk$
 - Treatment precluded in a lower level
- Services, **All:**
 - Goal directed therapy with rehabilitation potential ^(47, 48)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽⁴⁹⁾
 - Able to tolerate 2-3h/d of skilled therapy $\geq 5d/wk$ and ≥ 2 therapy disciplines
- Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽³⁸⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation

● ACUTE REHAB, Both:

- Level of care appropriateness, **Both:**
 - Medical practitioner, NP, PA assessment or oversight $\geq 3x/wk$
- Services, **All:**
 - Goal directed therapy with rehabilitation potential ^(47, 48)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽⁴⁹⁾
 - Able to tolerate $\geq 15h/wk$ of skilled therapy and ≥ 2 therapy disciplines ⁽⁵⁰⁾
- Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽³⁸⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation

● LONG TERM ACUTE CARE, Both:

- Level of care appropriateness, **Both:**
 - Treatment precluded in a lower level
- In lieu of acute or continued hospitalization or failed lower level of care, $\geq$ **One:**
 - Primary condition or illness and treatment of active comorbid condition(s) ⁽⁵¹⁾
 - Ventilator dependent $\geq 6h/d$ and weaning planned ^(52, 53)
 - Acute and comorbid conditions requiring prolonged hospitalization ⁽⁵⁴⁾
- Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽³⁸⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner

– Arrange transportation

Utilization Benchmarks

Condition or Procedure	LOS (days)	Type
811 RED BLOOD CELL DISORDERS WITH MCC	3.8	CMS GMLOS
812 RED BLOOD CELL DISORDERS WITHOUT MCC	2.8	CMS GMLOS

Notes:**1:**

Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

2:

Apheresis is a procedure that separates and removes elements in the blood (e.g., white blood cells (WBCs), platelets, stem cells, plasma bound organic compounds, or antibodies). Subsequent replacement with blood components, solvent detergent plasma (SDP), albumin, and/or normal saline may follow apheresis.

3:

Initial course refers to the initiation of a chronic hemodialysis/peritoneal dialysis program or initiation for acute renal pathology. A repeat initiation for a recurrence or exacerbation of an acute renal problem, provided that there was a gap of more than 14 days between treatments, would also qualify as an initial course.

4:

Immunosuppressant medications are used to treat conditions such as graft-versus-host disease (GVHD), hemolytic anemia, organ transplant, graft failure, rejection, inflammatory bowel disease, acute glomerulonephritis, and inflammatory cellulitis. Medications used for immunosuppression include prednisone, prednisolone, cyclosporine, azathioprine, mycophenolate mofetil, tacrolimus, sirolimus, everolimus, belatacept, antithymocyte globulin (ATG), and basiliximab.

5:

Hemodynamically stable patients who require treatment, assessment, or intervention every 2 to 4 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

6:

Hyperkalemia may cause electrocardiogram (ECG) changes such as a tall, narrow, T wave and a widening of the QRS complex (Sandau et al., Circulation 2017, 136: e273-e344).

7:

Dextrose 50%(0.50) with insulin is used for the treatment of hyperkalemia. This treatment shifts potassium intracellularly, and repeated doses can be given if the hyperkalemia persists. Other treatments that may be used simultaneously include potassium binding agents, diuretics, nebulized albuterol, calcium, sodium bicarbonate, and dialysis (Depret et al., Ann Intensive Care 2019, 9: 32).

8:

Oxygen therapy is the administration of oxygen at concentrations greater than ambient air (room air: 21%(0.21)) with the intent of treating and/or preventing the symptoms and manifestations of hypoxia. The oxygen concentration or percentage (FiO₂) delivered varies with the manufacturer's design, oxygen flow rate, the patient's respiratory rate, and tidal volume. Actual inspired FiO₂ with nasal cannula varies significantly with the patient's minute ventilation and pattern of breathing. For patients at risk for CO₂ retention, where a precise inspired FiO₂ is required, the Venturi mask is the preferred method. In general, patients who are very ill or have respiratory disease may require considerably higher flow rates to achieve the desired FiO₂. The following are estimates of O₂ delivered at the associated flow rates:

LOW-FLOW SYSTEMS**Nasal cannula**

Room air 21%(0.21)

1L 24%(0.24) **4L** 36%(0.36)

2L 28%(0.28) **5L** 40%(0.40)

3L 32%(0.32) 6L 44%(0.44)

Simple oxygen masks

- 35-50%(0.35-0.50) FiO₂ (5-10 L/min)
- Flow rates are usually maintained at 5 L/min or more to avoid accumulation of CO₂ in the mask

Venturi masks

- 24-50%(0.24-0.50) FiO₂ (4-12 L/min)

Partial rebreathing masks

- 40-70%(0.40-0.70) FiO₂ (6-10 L/min)

Non-rebreathing masks

- 60-80%(0.60-0.80) FiO₂ (minimum flow of 10 L/min)

HIGH-FLOW SYSTEMS**Air-entrainment masks/nebulizers**

- 24-40%(0.24-0.40) FiO₂

Heated, humidified high-flow concentrating nasal cannula (HFNC)

- 24-100%(0.24-1.0) FiO₂

9:

High-flow nasal cannula (HFNC) delivers a fraction of inspired oxygen (FiO₂) that is heated (up to 37 degrees Celsius) with 100%(1.0) relative humidity at flow rates up to 60 liters per minute, set independent from the FiO₂. The heated and humidified gas has been shown to improve comfort and aid in clearance of secretions. The high-flow rates reduce anatomical dead space by flushing out carbon dioxide from the upper airways, leading to improved work of breathing. In addition, HFNC reduces the amount of ambient air a patient can breathe in, resulting in less oxygen dilution. A Cochrane review was conducted in 2021 to determine the effectiveness of HFNC compared to standard oxygen therapy and noninvasive positive pressure ventilation (NIPPV) in adult patients requiring respiratory support in the intensive care unit (ICU). The systematic review found that HFNC had less treatment failure compared to standard oxygen therapy (i.e., face mask or nasal cannula with a flow rate of 15 L/min); however, it did not influence mortality rate or length of stay. When HFNC was compared to NIPPV, which included bilevel positive airway pressure (BiPAP) or continuous positive airway pressure (CPAP), there was no difference in treatment failure defined as intubation or re-intubation, mortality, or length of stay. In conclusion, HFNC has been shown to have less treatment failure compared to standard oxygen therapy and may be an alternative to NIPPV in patients treated for acute hypoxic respiratory failure (Lewis et al., Cochrane Database Syst Rev 2021, 3: Cd010172).

10:

Respiratory monitoring includes an evaluation of the patient's respiratory rate, heart rate, oxygen saturation, or work of breathing. Within 90 minutes of high-flow nasal cannula (HFNC) initiation, the patient's respiratory status should improve, indicating adequate respiratory support is being provided. If clinical improvement is not achieved within this time frame, additional respiratory support may be indicated (Lee et al., Intensive Care Med 2013, 39: 247-57).

11:

Hemodynamically unstable patients (or those with the potential to become unstable) who require treatment, assessment, or intervention every 1 to 2 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

12:

Hyperkalemia may be associated with weakness leading to muscle twitching, paresthesia, and paralysis.

13:

While no randomized, controlled trials have shown that continuous renal replacement therapy (CRRT) is superior

to intermittent dialysis with respect to survival, there are several advantages that may influence its use despite the lack of demonstrated survival benefit.

These advantages include:

- Gradual solute and fluid shifts resulting in greater hemodynamic stability, little risk of disequilibrium syndrome, and no worsening of brain edema
- Less need for fluid or nutritional restrictions, allowing for adequate nutrition without compromising fluid balance
- Decreased risk for intradialytic hypotension and recurrent kidney injury

Comprehensive guidelines recommend the use of CRRT for patients with hemodynamic instability, brain edema, persistent metabolic acidosis, and large fluid removal requirements. CRRT can be discontinued once renal recovery has been confirmed or another form of renal replacement is chosen based on the patient's clinical condition (Ostermann et al., *Kidney Int* 2020, 98: 294-309; Depret et al., *Ann Intensive Care* 2019, 9: 32; *Kidney Disease: Improving Global Outcomes (KDIGO)*, *Kidney International Supplements* 2012, 2: ii.-138).

14:

Fluid resuscitation involves the administration of repeated intravenous fluid boluses to acutely restore depleted intravascular volume. There is no evidence to support the use of colloids over crystalloids; studies have demonstrated that the primary choice for a resuscitation fluid is a balanced crystalloid solution, such as Lactated Ringer's (Myburgh, *N Engl J Med* 2018, 378: 862-3).

Bolus volumes for administration may be generally defined by the following:

- Fluid boluses of 250 to 1,000 milliliters in adults (Scales, *Br J Nurs* 2014, 23; Myburgh and Mythen, *N Engl J Med* 2013, 369: 1243-51; Vincent and De Backer, *N Engl J Med* 2013, 369: 1726-34)
- Fluid boluses of 10 to 20 milliliters per kilogram in pediatric patients. In some cases, boluses of 60 milliliters per hour or more may have to be administered in the first hour (Davis et al., *Crit Care Med* 2017, 45: 1061-93)

For patients with decreased cardiac reserve, significant renal dysfunction, or third-spacing, smaller volume boluses may be used (National Clinical Guideline Centre for Acute and Chronic Conditions, *IV Fluids in Children: Intravenous Fluid Therapy in Children and Young People in Hospital*. 2015. Reaffirmed 2020). The total number of fluid boluses required to restore intravascular volume and avoid fluid overload varies by patient (Joseph et al., *J Trauma Acute Care Surg* 2014, 76: 457-61; Holodinsky et al., *Crit Care* 2013, 17: R249; Kirkpatrick et al., *Intensive Care Med* 2013, 39: 1190-206). Dynamic measures such as bedside vena cava ultrasound and monitoring of urine output may assist in determining when volume resuscitation has been achieved (Kent et al., *J Surg Res* 2013, 184: 561-6; Zeng et al., *Am J Emerg Med* 2013, 31: 763-7; Weekes et al., *Acad Emerg Med* 2011, 18: 912-21).

15:

Refractory seizures, also known as uncontrolled, intractable, or drug-resistant seizures, are resistant to treatment and are poorly controlled with medication. Recurrent seizures are seizures that occur more than once and may occur without warning (Billington et al., *J Clin Med* 2016, 5:).

16:

A neurological assessment establishes a baseline so that subtle changes can be monitored. A comprehensive neurological assessment often includes an evaluation of:

- Mental status (e.g., level of consciousness, orientation, insight, calculation ability)
- Cranial nerves (e.g., olfactory, optic, vestibulocochlear)
- Motor system (e.g., muscle tone, strength, reflexes)
- Sensory system (e.g., light touch, pain, temperature)
- Coordination (e.g., orchestration and fluidity of movement)
- Gait (e.g., heel-to-toe straight-line walking)

17:

Glasgow Coma Scale (GCS) is a tool commonly used to measure neurological function. GCS evaluates the level of consciousness in acutely ill patients. A patient's GCS is the combined total score accumulated through the evaluation of the following 3 areas (ranging from 3-15) (Chou et al., In: *Glasgow Coma Scale for Field Triage of Trauma: A Systematic Review*. 2017):

- Eye opening - 4 = Spontaneous, 3 = To voice, 2 = To pain, 1 = None

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- Verbal response - 5 = Oriented, 4 = Confused, 3 = Inappropriate, 2 = Incomprehensible sounds, 1 = None
- Motor response - 6 = Obeys verbal commands, 5 = Localizes to pain, 4 = Withdraws to pain, 3 = Flexion response to pain, 2 = Extension response to pain, 1 = None

Of note, age may impact the relationship between anatomic injury severity and GCS scores such that the same GCS score may correspond to a greater severity of traumatic brain injury (TBI) in adults aged 65 and older than in those younger than 65 (Salottolo et al., JAMA Surg 2014, 149: 727-34). For patients less than 5 years of age, the verbal response component includes a modified scoring as follows:

- 5 = Appropriate word or social smile / Coos and babbles, 4 = Cries but consolable, 3 = Cries, screams, persistently irritable, 2 = Restless, agitated, moans, 1 = None

18:

Noninvasive positive pressure ventilation (NIPPV), also known as NIV, provides respiratory support by application of a tightly fitting facial mask, nasal mask, or helmet rather than an endotracheal tube or tracheostomy. In some cases, the use of NIPPV can avoid the need for endotracheal intubation and decreases the risk of barotrauma, lung injury, and/or infection. NIPPV is commonly delivered by a bilevel positive airway pressure ventilator (BiPAP), a continuous positive airway pressure device (CPAP), or a mechanical ventilator. Supplemental oxygen can be delivered at concentrations approximating 100%(1.0).

The decision to provide respiratory support via NIPPV, and the modality to provide NIPPV, is based upon the patient's specific clinical findings (e.g., medical condition leading to respiratory failure, underlying comorbidities, clinical progression, improvement). Examples of NIPPV devices are volumetric (i.e., deliver a defined volume), barometric (i.e., deliver a defined pressure), and combined (i.e., deliver defined volumes and pressure). The terminology for NIPPV delivery systems may differ between equipment manufacturers and provider organizations. InterQual® criteria do not differentiate between the different NIPPV modalities.

Whether or not high-flow nasal cannula (HFNC) should be classified as a component of NIPPV is unclear, and there is conflicting evidence in the medical literature. Where HFNC is an appropriate modality, InterQual® defines this in a specific criteria point. As such, NIPPV does not include HFNC (Hackett, A., PulmCCM 2018; Nardi et al., F1000Res 2017, 6: 290; Osadnik et al., Cochrane Database Syst Rev 2017, 7: CD004104; Allison and Winters, Emerg Med Clin North Am 2016, 34: 51-62; Gregoretti et al., Crit Care Clin 2015, 31: 435-57).

19:

Respiratory interventions include ventilator/noninvasive positive pressure ventilation (NIPPV) setting changes, chest physiotherapy, suctioning, and nebulizer or metered-dose inhaler (MDI) treatments.

20:

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

21:

When a criteria point states "within acceptable limits," it refers to either the patient's normal baseline, a newly established baseline, or parameters that the medical practitioner determines are acceptable.

22:

Oral intake should be at least equivalent to the patient's maintenance fluid requirements or within parameters which the medical practitioner deems appropriate. Maintenance fluid requirements should be increased in the presence of dehydration or insensible loss. Maintenance fluid rates are based on ideal body weight and reflect the minimum amount of fluid required to maintain hydration. Rates may be decreased in older patients or those with a history of renal impairment or heart failure (Scales, Br J Nurs 2014, 23).

23:

Relevant complications or comorbidities are conditions that require active management during an inpatient stay.

24:

Hypertension is defined as a systolic blood pressure (SBP) of greater than 130 mmHg or a diastolic blood pressure (DBP) of 80 mmHg or greater. Recommendations for initial management may include a conservative approach with

diet and lifestyle modification with or without medications. The goal of antihypertensive therapy is to reduce morbidity and mortality. In non-urgent situations, blood pressure (BP) control is achieved in the outpatient setting. Goals for target BP level should be individualized, but a SBP less than 130 mmHg and a DBP less than 80 mmHg are desirable. Populations at high risk include patients with the following conditions (Whelton et al., Hypertension 2018, 71: 1269-324):

- Clinical cardiovascular disease or 10-year atherosclerotic cardiovascular disease (ASCVD) risk 10%(0.1)
- Heart failure
- Stable ischemic heart disease
- Chronic kidney disease
- Chronic kidney disease after renal transplantation

25:

Guidelines recommend preprandial glucose targets of 80-130 mg/dL (4.4-7.2 mmol/L) and a random blood glucose target range of 100-180 mg/dL (5.6-10.0 mmol/L) in hospitalized patients. In stable patients with previous tight glycemic control, more stringent targets may be appropriate. In patients with severe comorbidities, a history of hypoglycemia, terminal illness, late onset diabetes, and in older patients, less stringent targets may be appropriate. The preferred method for achieving and maintaining glucose control in non-critically ill patients is scheduled subcutaneous insulin administration. Guidelines discourage the sole use of sliding scale insulin in the inpatient hospital setting. Rather, an insulin regimen consisting of basal, prandial, and correctional insulin is recommended in hospitalized patients (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

26:

Instruction: Insulin adjustments should be based on blood glucose values obtained by lab draw or glucose monitor. Intermittent insulin must be administered at least 3 times in 24 hours. If the patient is on a continuous insulin infusion, the rate must be adjusted at least twice in 24 hours. A dose that is held based on a normal or low blood glucose value should be included in the count of doses administered within the 24-hour period.

27:

A fever is suspected as being new onset when a patient has been afebrile and then develops a new fever or when the fever pattern changes.

28:

The Centers for Disease Control and Prevention (CDC) define a fever as greater than or equal to 100.4 degrees Fahrenheit (i.e., greater than or equal to 38 degrees Celsius) and do not specify the route of measurement (Centers for Disease Control and Prevention, Definitions of symptoms for reportable illness. 2017). The route utilized may be directed by institutional protocols, equipment availability, patient preference, or other patient-specific factors.

29:

A drug-induced fever is a non-infectious febrile response following the administration of a drug that resolves when the drug is discontinued. The patient is often well-appearing and demonstrates resolution of the underlying disease process, despite the presence of a fever. A drug-induced fever should be suspected when the cause of the fever remains unknown, despite a full diagnostic work-up, or when the patient is receiving a drug frequently associated with fever. The most common agents triggering fever include anticonvulsants, anti-infectives, cardiac medications, allopurinol, and heparin.

The following drugs are also capable of inducing hyperthermia (Walter and Carraretto, Journal of the Intensive Care Society 2015):

- Serotonin-noradrenalin reuptake inhibitors
- Selective serotonin reuptake inhibitors
- Monoamine oxidase inhibitors
- Tricyclic antidepressants
- Bupropion
- Opioids
- Central nervous system stimulants
- Anticholinergics
- Antipsychotics
- Antispasmodics

- Sympathomimetic agents

Treatment involves discontinuing medications that are unnecessary, recently initiated, or highly associated with inducing fever. Resolution of the fever should occur within 48 to 72 hours of discontinuing the precipitating agent (Patel and Gallagher, Pharmacotherapy 2010, 30: 57-69; Eyer and Zilker, Crit Care 2007, 11: 236).

30:

In patients with no underlying heart disease, occasional, nonconsecutive premature ventricular contractions (PVCs) less than 6/min are common and not considered a concerning finding. Three or more consecutive PVCs at a rate of 100 bpm or more that terminate spontaneously within 30 seconds are considered nonsustained ventricular tachycardia (Marine, Card Electrophysiol Clin 2016, 8: 525-43; Buxton et al., Circulation 2006, 114: 2534-70).

31:

Following an initial 12-lead electrocardiogram (ECG) demonstrating no abnormal findings in the setting of hyperkalemia or hypokalemia, continued monitoring may be accomplished by telemetry (Bridgeman et al., Nephrol Dial Transplant 2019, 34: iii45-iii50).

32:

Instruction: This line of criteria is intended for the patient receiving condition-directed maintenance IV fluid and excludes the patient who has IV fluid ordered to keep vein open (KVO).

33:

Examples of potassium binding agents include but are not limited to:

- Sodium polystyrene sulfonate (Kayexalate® or Kionex®)
- Sodium zirconium cyclosilicate (Lokelma®)
- Patiromer (Veltassa®)

34:

Baseline refers to either the patient's normal baseline or a newly established baseline. In the absence of documentation, a patient's baseline status may be presumed to be normal.

35:

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

36:

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

37:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

38:

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Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

39:

Close follow-up of older adults, pediatric, or at-risk patients after discharge can help minimize hospital readmission and total health care costs (Rising et al., Ann Emerg Med 2013, 62: 145-50; Frei-Jones et al., Pediatr Blood Cancer 2009; 52(4): 481-485; Kripalani et al., JAMA 2007; 297(8): 831-841). Studies have shown that for greater than 50% (0.50) of patients re-hospitalized within 30 days of discharge, there was no visit to a physician's office between the time of discharge and re-hospitalization (Hernandez et al., JAMA 2010, 303: 1716-22; Jencks et al., N Engl J Med 2009; 360(14): 1418-1428). Efforts to reduce readmissions should focus on patients known to be at high risk and patients who have frequent visits to the emergency department (ED). Patients at high risk may have multiple ED visits prior to readmission (Rising et al., Ann Emerg Med 2013, 62: 145-50).

40:

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk
- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home would be cumbersome and places the patient at risk
- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

41:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

42:

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor

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patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

43:

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

44:

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.

Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.

Evaluation and Treatment

Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit
- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

45:

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

46:

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems can be identified by the clinical assessment. The clinical assessment includes:

- History and physical exam (e.g., vital signs, height, weight)
- Pain assessment includes cause, intensity, quality, onset, duration, and effects on quality of life (e.g.,

activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal relationships). Pain relievers should also be included

- Nutritional and hydration status
- Functional ability
- Safety and infection control measures
- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects, and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

47:

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

48:

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

49:

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.

The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations
- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

50:

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility.

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Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

51:

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

52:

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

53:

Notable causes for prolonged mechanical ventilation include:

- Ventilator-acquired pneumonia
- Renal failure
- Heart failure
- Exacerbation of chronic obstructive pulmonary disease (COPD)
- Chest wall disorder
- Neuromuscular diseases
- Traumatic brain injury
- Sepsis
- Complications following cardiac surgical procedures

Major contributing factors for failure to wean include (Ambrosino and Vitacca, Multidiscip Respir Med 2018, 13: 6; Davies et al., Br J Anaesth 2017, 118: 563-9):

- Increased work of breathing
- Impaired respiratory drive
- Inspiratory muscle weakness

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Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784). Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.